Implementation of Ping Program.

Ex: 13
17/9/2026

Aim:

To write a Java Program to Ping a hostname or IP address using the InetAddress class and Check whether the host is reachable, also work response time.

Procedure.

1. Open a gedit (or) as terminal and Create a file named mypingjava
2. Import InetAddress and scanner classes
3. Get the hostname or IP address.
4. Use InetAddress.getByName() to obtain the IP address
5. Use isReachable(3000) to check whether the host is reachable with 3 seconds.
6. Calculate and display the response time in milliseconds
7. Repeat the Ping Limit with 1 second delay Also each attempt

Result:

The Java program was successfully executed to ping given hostname / IP Address It displays whether the host is reach the and

Ex: 13 Implement

Code:

```
import socket
import struct
import time
import os

def checksum(data):
    sum = 0
    for i in range(0, len(data)-1, 2):
        sum += (data[i] << 8) + data[i+1]
    if len(data) % 2:
        sum += data[-1] << 8
    sum = (sum >> 16) + (sum & 0xffff)
    sum += sum >> 16
    return ~sum & 0xffff

def ping(host):
    try:
        dest = socket.gethostbyname(host)
        sock = socket.socket(socket.AF_INET,
        socket.SOCK_RAW, socket.IPPROTO_ICMP
    socket.settimeout(2)
    packet_id = os.getpid() & 0xffff
    sequence = 1
    header = struct.pack("!BBHHH", 8, 0, 0, packet_id,
    data = struct.pack("!d", time, sequence)
    checksum = checksum(header + data)
    header = struct.pack(
```